

MAE 184 Flight Simulation Techniques  
Homework #1 Due WED 21 APR 2025

Please submit final answers using the formatted "Quiz" on the Canvas course website. This worksheet is for reference only.

Problem 1.1 (2 pts)

A flight test technique used to estimate the drag coefficient of early airships involved accelerating the vehicle up to its maximum forward speed, cutting the engines, and then measuring how the vehicle speed slowed over time. Use Euler integration to find the vehicle speed after it has coasted without engine power for 30 seconds. Use an integration time step of 0.1 seconds and assume an initial airspeed of 70 ft/s.

$$m \frac{dV}{dt} = -\frac{1}{2} \rho V^2 \nabla^{2/3} C_D$$

mass,  $m = 200$  slugs

density,  $\rho = 0.002377$  slug/ft<sup>3</sup>

airship volume,  $\nabla = 89,300$  ft<sup>3</sup>

drag coefficient,  $C_D = 0.044$

airship velocity after coasting 30 seconds = \_\_\_\_\_ [ft/s]

Problem 1.2 (2 pts)

Two Helios spacecraft were deployed in the mid 1970's to study solar processes of the sun. Each spacecraft had a mass of about 370 kg and both were designed to spin at a constant 60 rev/min about the 3-axis. Simulate the spacecraft's initial condition response with 1-axis rotation rate ( $\omega_1$ ) of 0.1 rad/s and the 2-axis rotational rate ( $\omega_2$ ) of 0.01 rad/s. Use the Ralston integration method with step size of 0.05 seconds and run the simulation for a total elapsed time of 30 seconds. Submit an image plot (PNG, BMP, JPG) with 1-axis rotation rate (units: deg/s) on the x-axis and the 2-axis rotational rate (units: deg) on the y-axis ordinate.

$$I_1 = 182 \text{ kg.m}^2 \quad I_2 = 187 \text{ kg.m}^2 \quad I_3 = 194 \text{ kg.m}^2$$

$$I_1 \frac{d\omega_1}{dt} = I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_3 \omega_2$$

$$I_2 \frac{d\omega_2}{dt} = I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1$$

$$I_3 \frac{d\omega_3}{dt} = I_3 \dot{\omega}_3 = 0$$

Problem 1.3 (2 pts)

Alaska Airlines recently introduced non-stop flights between San Diego CA (KSAN) and Honolulu HI (PHNL). Estimate the total distance between KSAN and PHNL using two different modes of navigation. The first mode assumes the airplane travels along a rhumb line path between each of the waypoints listed below. The second navigation mode assumes the pilot ignores the intermediate waypoints and follows a great circle path between KSAN and PHNL. Report as your answer the magnitude of the difference between the total trip distance computed by each mode of navigation. Ignore aircraft altitude and assume a spherical Earth with semi-major axis equal to 6378137 m.

| Name  | Latitude (deg) | Longitude (deg) |
|-------|----------------|-----------------|
| KSAN  | 32.7335556     | -117.1896667    |
| ETECO | 30.2858333     | -131.6361111    |
| DIALO | 28.7016667     | -139.4266667    |
| DUSAC | 26.0061111     | -146.1675000    |
| DRAYK | 23.0241667     | -152.5777778    |
| PHNL  | 21.3178275     | -157.9202627    |

magnitude of trip distance difference = \_\_\_\_\_ [km]

Problem 1.4 (2 pts)

A private pilot plans to fly a rhumb line course from San Diego airport (KSAN) to San Francisco airport (KSFO). If there is no wind, the pilot expects the trip to take about 130 minutes at a airspeed of 180 nmi/hr. However, when the actual flight occurs, the wind is blowing (from west) toward east at 20 nmi/hr. How long will the flight actually take if the pilot maintains the 180 nmi/hr airspeed? (1 nmi = 6076 ft)

| Name | Latitude (deg) | Longitude (deg) |
|------|----------------|-----------------|
| KSAN | 32.7336        | -117.1897       |
| KSFO | 37.6167        | -122.3750       |

- |                |                |
|----------------|----------------|
| a. 120 minutes | c. 140 minutes |
| b. 130 minutes | d. 160 minutes |

Problem 1.5 (2 pts)

Estimate the ratio of atmospheric density at Coors Field in Denver CO (altitude = 5280 ft), on a non-standard "hot" day, relative to the sea-level or base level for the same day.

density ratio =  $\rho/\rho_0 = \rho(5280)/\rho(0)$

- |         |         |
|---------|---------|
| a. 0.83 | c. 1.17 |
| b. 0.86 | d. 1.20 |